User Manual: Personal Information Management System

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1. What the program does

The Personal Information Management (PIM) program keeps four kinds of personal information record (PIR): notes, tasks, events, and contacts. You can create, list, inspect, change, search, and delete them at a command prompt. You can save the whole collection to a `.pim` file and load it in a later run. The program is local and uses text commands; an event's alarm time is stored information, not an active notification.

Each new PIR receives a positive integer ID. IDs start at 1 in a new empty process, increase after successful creation, and are not reused after deletion. A record's ID and type cannot be changed. Duplicate field values are allowed. A failed creation does not consume an ID.

2. Start, prompt, and exit

Extract the delivery ZIP, then open PowerShell inside `03_Implementation`, the folder containing `src` and the two manuals. If working from a repository checkout, use its root instead. Run:

```
python src/main.py
```

The program writes `pim>` and waits for one command per line. Type `help` for the command summary, or `help search` (and any other command name) for usage, details and an example. Type `exit` to finish. End-of-file also exits normally. If the collection has unsaved changes, exit or EOF prints one warning and still terminates; it does not wait for another command. Exiting does **not** save automatically: run `save` first if you want to keep the current collection for a later process. Every new process starts empty until you load a saved file.

Ctrl+C terminates without a traceback (process exit code 130). If a program receiving piped output closes its input, the PIM exits quietly. Neither case saves the collection automatically.

The Windows 11 development and verification versions are Python 3.11.9, 3.12.3 and 3.13.5. The Developer Manual provides setup and test commands; `Test_Coverage.pdf` identifies the exact verification results for this delivery.

3. Input rules and record fields

Commands, type names, and field names ignore letter case. Separate command arguments with whitespace. Enclose a value or path containing spaces in double quotes; a one-token non-search value can be unquoted. Inside a quoted argument, `\"` represents a literal double quote and `\\` represents a literal backslash. Other backslash escapes are invalid. A quoted argument must be followed by whitespace or the end of the command. Search expressions have their own quoting rules in §5.

All record fields are required. Text fields must be single-line valid UTF-8 and nonempty after trimming; the trimmed text is stored. A mobile number is text, so `+` and leading zeroes are retained; the program does not apply a regional phone-number format. Time fields accept no surrounding whitespace and use exact local 24-hour `YYYY-MM-DD HH:mm` at minute precision, for example `2026-10-01 09:00`. Impossible calendar dates and other time formats are rejected. Past dates are allowed, and an event's `alarm_time` need not precede its `start_time`.

Type	add **field order**	**Fields shown by** show
note	text	text
task	description, deadline	description, deadline
event	description, start_time, alarm_time	description, start_time, alarm_time
contact	name, address, mobile_number	name, address, mobile_number

4. Command reference

`TYPE` is one of the four types above. `ID` is a positive decimal integer. `FIELD` is a data field belonging to that record's type. Quoted forms below also work for one-token values.

Command	Format and example	Result
Help	`help` or `help search`	Prints the overview or a command-specific guide; topics ignore case.
Create note	`add note "memo"`	Creates a note; prints `Created note ID`.
Create task	`add task "draft" "2026-10-01 09:00"`	Creates a task; prints `Created task ID`.
Create event	`add event "meeting" "2026-10-02 10:00" "2026-10-01 08:00"`	Creates an event; prints `Created event ID`.
Create contact	`add contact "Li" "Room 1" "+00123"`	Creates a contact; prints `Created contact ID`.
List all or one type	`list` or `list task`	Prints one summary per PIR in ascending ID order. A type argument filters the list.
Show one or all	`show 2` or `show all`	Prints the ID, type, and every data field. `show all` is in ascending ID order.
Change one field	`update 2 description "revised"`	Replaces one field on an existing PIR; prints `Updated 2`. Use the exact field name for that PIR type.
Delete one	`delete 2`	Removes the specified PIR; prints `Deleted 2`. Its ID is not reused.
Search	`search type = "task"`	Prints matching summaries in ascending ID order. Full grammar is in §5.
Save	`save "snapshot.pim"`	Writes the full collection and next-ID counter; prints `Saved N PIR(s) to PATH`.
Load	`load "snapshot.pim"`	Loads a validated replacement when there are no unsaved changes. Append `force=true` to explicitly discard unsaved changes.
Exit	`exit`	Ends the process without an implicit save.

`exit` takes no arguments; `help` accepts at most one command topic. `add` needs exactly the fields listed in the table and in that order. `update` changes exactly one field per command. `list` accepts at most one type. `show`, `delete` and `save` each require exactly one argument. `load` requires a path and optionally the final token `force=true` or `force=false`; the option ignores case. `search` requires an expression after the command word. Blank input simply returns to the next prompt.

Protecting unsaved changes

A successful create, update or delete marks the collection as unsaved. Saving or loading successfully clears that status. Failed operations and reading/listing/searching do not clear it. An update that repeats the same value still marks the session unsaved.

If you have unsaved edits, `load "snapshot.pim"` reports an error and leaves the current data intact. Save those edits first, or deliberately use `load "snapshot.pim" force=true` to discard them and load the snapshot. `force=false` keeps protection. A failed forced load still preserves the existing records, next ID and unsaved status. Exit and EOF only warn; save before exiting to retain your work.

Finding input errors

Search errors append `at character N`, counting from the first character of EXPR after `search` and its separator. For example, `search type =` points to character 7, where the missing literal should begin. A missing item at the end points to the expression length plus one. Non-ASCII characters count as single Unicode code points; tabs are not expanded into screen columns. Malformed quoted command arguments use positions in the whole command line, including leading spaces. Correct the indicated input and try again, or use `help search`. Errors return to the prompt.

Reading output

Terminal control characters such as Escape, NUL and backspace are displayed as visible `\x1b`, `\x00` and `\x08` text. Normal line breaks and tabs remain layout characters. If the output encoding cannot display a character, it appears as a Python-style escape such as `\u4e2d`; a UTF-8 terminal displays ordinary Chinese and emoji directly. These are display representations only: searching and `.pim` files retain the original stored text. These output escapes do not add new command-input escape syntax.

`list` and `search` display `ID | type | first text field` on each line. The first text field is `text` for a note, `description` for a task or event, and `name` for a contact; the entire stored value is shown. `show` prints `id:` and `type:` followed by all data-field labels and values. `show all` separates records with a blank line. An empty `list` or `show all` prints `No PIRs.`; an empty search prints `No matching PIRs.`. These messages are different so you can distinguish an empty collection/list filter from a search with no result.

5. Search expressions

A `search` command defines and applies its criterion at once. Every search value, **including a one-word type or text value**, must be in double quotes. Quoted search literals support only `\"` and `\\` escapes. Search field names, `contains`, and type names ignore letter case. Text containment also ignores letter case, matches a substring, and requires a nonempty quoted string. Time comparisons use the exact local time format from §3 and compare chronological values.

Criterion	Valid form	Example
PIR type	`type = "TYPE"`	`type = "contact"`
Text containment	`TEXT_FIELD contains "STRING"`	`address contains "Room"`
Time comparison	`TIME_FIELD < "TIME"`, `>`, or `=`	`deadline < "2026-10-02 00:00"`

`TEXT_FIELD` is one of `text`, `description`, `name`, `address`, or `mobile_number`. `TIME_FIELD` is one of `deadline`, `start_time`, or `alarm_time`. Every time field supports each of `<`, `>`, and `=`. A valid field absent from a particular PIR evaluates false for that PIR; negating that condition evaluates true. An unknown field is an error for the whole search.

Combine complete criteria with `&&` (AND), `||` (OR), `!` (NOT), and parentheses. Parentheses bind first, then `!`, then `&&`, then `||`; operators at the same precedence associate left to right. Spaces around symbolic operators and parentheses are optional when the token boundary is clear. For example:

```
search (type = "task" && deadline < "2026-10-02 00:00") || text contains "memo"
search !name contains "Li"
search start_time = "2026-10-02 10:00"
```

The first line is demonstrated in §8. The other forms use the same search grammar. A malformed expression, wrong operator, invalid type or time value, empty `contains` string, or unmatched parenthesis produces `Error: SEARCH: ...` and no partial search result.

6. Saving, loading, and paths

Use a filename ending in the exact lowercase suffix `.pim`; `.PIM` and other suffixes are rejected. Quote a path containing spaces. A relative path is interpreted against the working directory from which this program was started; the success message shows the path used. An absolute path is used as entered. The parent directory must already exist.

On Windows, forward slashes avoid escape ambiguity: for example, `save "my records/snapshot.pim"` after creating the `my records` folder. If using backslashes inside quotes, double each one: `save "my records\\snapshot.pim"`. Both forms select the same file. Apply the same path rules to `load`.

`save` writes all current PIRs, including an empty collection, and the next-ID counter as a UTF-8 version-one `.pim` snapshot. Saving to an existing destination replaces that file after a complete temporary write. A failed save does not change the in-memory records; if the destination already exists, the failed write leaves its prior bytes unchanged under ordinary handled I/O failure. Save does not happen automatically when exiting.

`load` reads and validates the **whole** `.pim` file before replacing the current collection. It replaces rather than merges records. If it fails, the active collection and next-ID counter remain intact. After a successful load, a new PIR uses the higher of the session's former next ID and the file's saved next ID. This prevents reuse of an ID already issued in the current session, even when loading an older snapshot. A fresh process only knows the counter present in the loaded file.

The `.pim` file is a program snapshot, not a free-form text import. The loader rejects unreadable files, invalid UTF-8 or JSON (including excessively nested malformed data), unsupported format/version, duplicate or unexpected keys, invalid fields or values, duplicate or unordered IDs, and an invalid next-ID counter. A rejected file produces a handled error so you can enter another command. Editing the file manually can make it unloadable.

7. Invalid input and recovery

Handled errors use `Error: CATEGORY: DETAIL` and then show another `pim>` prompt. The detail identifies the rejected input or failure and may include an operating-system file error. Correct the command and retry; a handled failure does not partially change the in-memory PIR collection.

Category	Typical cause	What to check
`COMMAND`	Unknown command, wrong argument count, malformed quote/escape, or a nonpositive/nondigit/overlong ID token	Command form, spaces, quotes, and positive decimal ID.
`VALIDATION`	Unknown PIR type, empty or multiline field, invalid field for `update`, or impossible/wrongly formatted time	Field order and spelling, required text, and `YYYY-MM-DD HH:mm`.
`NOT_FOUND`	A valid positive ID is absent	Use `list` to see current IDs.
`SEARCH`	Missing or malformed criterion, invalid search field/type/operator/literal	Quote each literal and check the grammar in §5.
`PATH`	Path lacks exact lowercase `.pim` suffix	Use a `.pim` filename.
`FORMAT`	Existing `.pim` content is invalid or unsupported	Load a snapshot written by this program.
`IO`	File is missing, directory is unavailable, or the operating system refuses a read/write	Check the resolved path and filesystem access.

The CLI tests check that an invalid update leaves the old value visible, a failed load leaves the active collection visible, and the next command still runs. The independent review reproduced `Error: VALIDATION: text must not be empty` for `add note ""`, `Error: COMMAND: invalid positive decimal ID: '0'` for `show 0`, and `Error: NOT_FOUND: record ID 99 was not found` for `show 99`. Exact operating-system wording after an `IO` error depends on the path and machine.

8. Complete two-process example

The usability release verification reran this example on 2026-10-23. Both Python processes exited with code 0, the snapshot existed, and the output matched the transcript below. Only the absolute temporary directory is replaced with `<TEMP_DIR>` below. That token is a redaction of the test location, not text to type as a path. Run both sessions from the **same** working directory so the relative `snapshot.pim` resolves to the same file.

First start `python src/main.py`, then enter:

```
add note "memo"
add task "draft" "2026-10-01 09:00"
add event "meeting" "2026-10-02 10:00" "2026-10-01 08:00"
add contact "Li" "Room 1" "+00123"
list
search (type = "task" && deadline < "2026-10-02 00:00") || text contains "memo"
update 2 description "revised"
delete 1
save "snapshot.pim"
exit
```

Recorded output:

```
pim> Created note 1
pim> Created task 2
pim> Created event 3
pim> Created contact 4
pim> 1 | note | memo
2 | task | draft
3 | event | meeting
4 | contact | Li
pim> 1 | note | memo
2 | task | draft
pim> Updated 2
pim> Deleted 1
pim> Saved 3 PIR(s) to <TEMP_DIR>\snapshot.pim
pim>
```

Start `python src/main.py` again from the same directory and enter:

```
load "snapshot.pim"
show all
add note "after load"
show 5
exit
```

Recorded output:

```
pim> Loaded 3 PIR(s) from <TEMP_DIR>\snapshot.pim
pim> id: 2
type: task
description: revised
deadline: 2026-10-01 09:00

id: 3
type: event
description: meeting
start_time: 2026-10-02 10:00
alarm_time: 2026-10-01 08:00

id: 4
type: contact
name: Li
address: Room 1
mobile_number: +00123
pim> Created note 5
pim> id: 5
type: note
text: after load
pim> Warning: unsaved changes will be lost; use save "PATH.pim" before exiting.
```

The loaded collection contains IDs 2–4 because ID 1 was deleted before saving. The next creation gets ID 5, confirming that deletion and reload did not reuse ID 1.

The final warning is expected: note 5 was added after loading and was not saved. It is absent from the snapshot on disk.